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MAST20030: Differential Equations
Tutorial Solutions: Week 3

Solution to Question 1.

(a) Letting $x'(t) = v(t)$ gives $v'(t) = 2x(t) + v(t)$. Putting this into matrix form gives

$$\boldsymbol{y}'(t) = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \boldsymbol{y}(t)$$

(b) Substituting the trial solution into the vector-valued ODE gives

$$\lambda \boldsymbol{v} \mathrm{e}^{\lambda t} = \boldsymbol{A}\boldsymbol{v}\mathrm{e}^{\lambda t} \qquad \Rightarrow \qquad \lambda \boldsymbol{v} = \boldsymbol{A}\boldsymbol{v}$$

and hence λ and $\boldsymbol{v}$ are the eigenvalues and eigenvectors of $\boldsymbol{A}$ respectively.

(c) The eigenvalues of $\boldsymbol{A}$ are given by

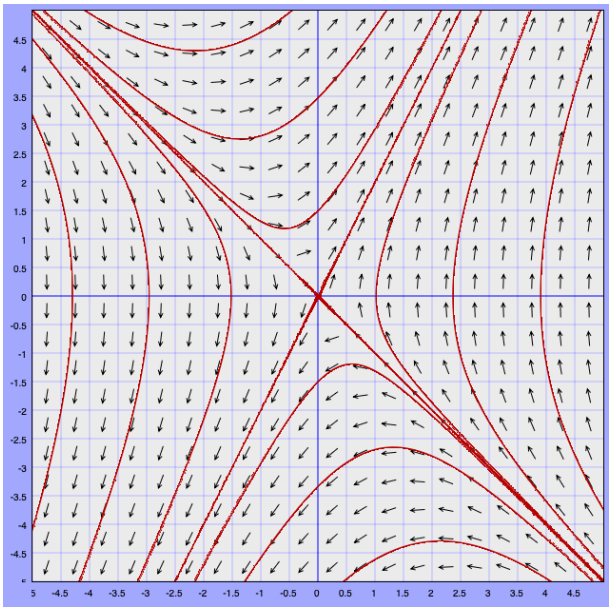
$$\det\left[\begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} - \lambda \boldsymbol{I}\right] = \lambda^2 - \lambda - 2 = 0 \qquad \Rightarrow \qquad \lambda_1 = -1 \qquad \text{and} \qquad \lambda_2 = 2$$

The eigenvectors are therefore

$$\lambda_1 = -1\text{:} \qquad \boldsymbol{A} - (-1)\boldsymbol{I} = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{A} + \boldsymbol{I} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\lambda_2 = 2\text{:} \qquad \boldsymbol{A} - 2\boldsymbol{I} = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{A} - 2\boldsymbol{I} = \begin{pmatrix} -2 & 1 \\ 0 & 0 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

(d) The phase portrait is



Trajectories asymptote to the line spanned by $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ as $t \to -\infty$ and they asymptote to the line spanned by $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ as $t \to \infty$.

Solution to Question 2.

(a) The vector-valued ODE is

$$\boldsymbol{y}'(t) = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix} \boldsymbol{y}(t) + \begin{pmatrix} 0 \\ \mathrm{e}^{-t} \end{pmatrix}$$

(b) We have that $\boldsymbol{D} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$, $\boldsymbol{S} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$ and $\boldsymbol{S}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$

(c) We let $\boldsymbol{z}(t) = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \boldsymbol{y}(t)$ and so we have that

$$\boldsymbol{z}'(t) = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \boldsymbol{z}(t) + \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ \mathrm{e}^{-t} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix} \boldsymbol{z}(t) + \frac{1}{3}\begin{pmatrix} -\mathrm{e}^{-t} \\ \mathrm{e}^{-t} \end{pmatrix}$$

and so if $\boldsymbol{z}(t) = \begin{pmatrix} z_1(t) \\ z_2(t) \end{pmatrix}$ then $z_1'(t) = -z_1(t) - \frac{1}{3}\mathrm{e}^{-t}$ and $z_2'(t) = 2z_2(t) + \frac{1}{3}\mathrm{e}^{-t}$.

(d) We can solve the two ODEs with an integrating factor to get $z_1(t) = C_1\mathrm{e}^{-t} - \frac{1}{3}t\mathrm{e}^{-t}$ and $z_2(t) = C_2\mathrm{e}^{2t} - \frac{1}{9}\mathrm{e}^{-t}$, i.e.,

$$\boldsymbol{z}(t) = \begin{pmatrix} C_1\mathrm{e}^{-t} - \frac{1}{3}t\mathrm{e}^{-t} \\ C_2\mathrm{e}^{2t} - \frac{1}{9}\mathrm{e}^{-t} \end{pmatrix}.$$

Because $\boldsymbol{z}(t) = \boldsymbol{S}^{-1}\boldsymbol{y}(t)$ we just have to multiply $\boldsymbol{z}(t)$ by $\boldsymbol{S}$ to get the final answer,

$$\boldsymbol{y}(t) = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} C_1\mathrm{e}^{-t} - \frac{1}{3}t\mathrm{e}^{-t} \\ C_2\mathrm{e}^{2t} - \frac{1}{9}\mathrm{e}^{-t} \end{pmatrix}$$

Solution to Question 3.

(a) The vector-valued ODE is

$$\boldsymbol{y}'(t) = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix} \boldsymbol{y}(t)$$

(b) The eigenvalues of $\begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}$ are

$$\det\left[\begin{pmatrix} -\lambda & 1 \\ -4 & -\lambda \end{pmatrix}\right] = \lambda^2 + 4 = 0 \qquad \Rightarrow \qquad \lambda = \pm 2i$$

To find the eigenvectors

$$\begin{pmatrix} \mp 2i & 1 \\ -4 & \mp 2i \end{pmatrix} \boldsymbol{v} = \boldsymbol{0} \qquad \Rightarrow \qquad \boldsymbol{v} = \begin{pmatrix} \mp i \\ 2 \end{pmatrix}$$

and so the general solution is

$$\boldsymbol{y}(t) = C_1 \begin{pmatrix} -i \\ 2 \end{pmatrix} \mathrm{e}^{2it} + C_2 \begin{pmatrix} i \\ 2 \end{pmatrix} \mathrm{e}^{-2it}$$

(c) We have that $x(t) = -C_1 i\mathrm{e}^{2it} + C_2 i\mathrm{e}^{-2it}$ and so $x^2(t) = -C_1^2\mathrm{e}^{4it} + 2C_1C_2 - C_2^2\mathrm{e}^{-4it}$. Similarly $v^2(t) = 4C_1^2\mathrm{e}^{4it} + 8C_1C_2 + 4C_2^2\mathrm{e}^{-4it}$. We can eliminate the t dependence by taking the combination $4x^2(t) + v^2(t)$, i.e.,

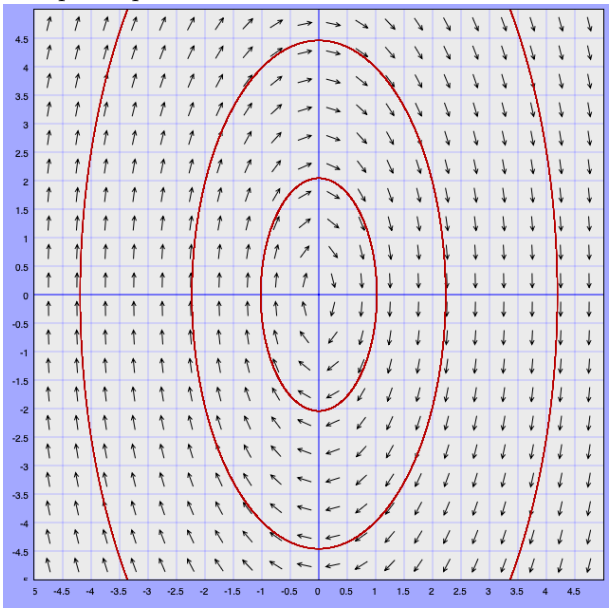
$$4x^2(t) + v^2(t) = -4C_1^2\mathrm{e}^{4it} + 8C_1C_2 - 4C_2^2\mathrm{e}^{-4it} + 4C_1^2\mathrm{e}^{4it} + 8C_1C_2 + 4C_2^2\mathrm{e}^{-4it} = 16C_1C_2$$

which upon rearranging gives

$$x^2(t) + \frac{v^2(t)}{2^2} = C_0 \qquad \text{where} \qquad C_0 = 4C_1C_2.$$

This tells us that the trajectories on the phase portrait will be ellipses!

(d) The phase portrait is



We can determine the direction of the trajectories by examining the direction of the rate of change vector for specific phase-points. For example, consider the point $\begin{pmatrix} x \\ v \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $\boldsymbol{y}' = \begin{pmatrix} 0 & 1 \\ -4 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix}$, i.e., the rate of change vector points directly downwards along the positive x-axis and so the trajectories must be clockwise.

Solution to Question 4.

(a) This follows from $\mathbf{A}^n = (\mathbf{S}\mathbf{D}\mathbf{S}^{-1})\cdots(\mathbf{S}\mathbf{D}\mathbf{S}^{-1}) = \mathbf{S}\mathbf{D}^n\mathbf{S}^{-1}$ and a straightforward computation using the definition of the matrix exponential in terms of a series,

$$\mathrm{e}^{\mathbf{A}} = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!} = \mathbf{S}\sum_{n=0}^{\infty} \frac{\mathbf{D}^n}{n!}\mathbf{S}^{-1} = \mathbf{S}\mathrm{e}^{\mathbf{D}}\mathbf{S}^{-1} .$$

(b) We have $\mathbf{D}^n = \mathrm{diag}(\lambda_1^n, \ldots, \lambda_N^n)$ and hence

$$\mathrm{e}^{\mathbf{D}} = \sum_{n=0}^{\infty} \frac{\mathbf{D}^n}{n!} = \sum_{n=0}^{\infty} \frac{1}{n!}\mathrm{diag}(\lambda_1^n, \ldots, \lambda_N^n) = \mathrm{diag}(e_1^{\lambda}, \ldots, e^{\lambda_N}) .$$

Solution to Question 5.

(a) The eigenvalues of $\mathbf{A}$ are -1 and 2. Corresponding eigenvectors are

$$\lambda = -1 : \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \lambda = 2 : \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

(b) Let

$$\mathbf{D} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad \mathbf{S}^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}$$

Then

$$\begin{aligned} e^{-\mathbf{A}t} = \mathbf{S}e^{-\mathbf{D}t}\mathbf{S}^{-1} &= \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} e^t & 0 \\ 0 & e^{-2t} \end{pmatrix}\frac{1}{3}\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{1}{3}\begin{pmatrix} 2e^t + e^{-2t} & -e^t + e^{-2t} \\ -2e^t + 2e^{-2t} & e^t + 2e^{-2t} \end{pmatrix}. \end{aligned}$$

The solution to (1) is therefore

$$\mathbf{r}(t) = e^{-\mathbf{A}t}\mathbf{r}(0) = \frac{1}{3}\begin{pmatrix} 2e^t + e^{-2t} & -e^t + e^{-2t} \\ -2e^t + 2e^{-2t} & e^t + 2e^{-2t} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2e^t + e^{-2t} \\ -2e^t + 2e^{-2t} \end{pmatrix}.$$

(c) Changing the initial conditions is straightforward:

$$\mathbf{r}(t) = e^{-\mathbf{A}t}\mathbf{r}(0) = \frac{1}{3}\begin{pmatrix} 2e^t + e^{-2t} & -e^t + e^{-2t} \\ -2e^t + 2e^{-2t} & e^t + 2e^{-2t} \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} -e^t + e^{-2t} \\ e^t + 2e^{-2t} \end{pmatrix}.$$